An Exact Rogers–Fine Proof that the Cyclotomic Mock-Theta Constant Satisfies $c_0(T_p) = 3$

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with a companion Lean 4 formalization

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Abstract

Let $p \geq 7$ be prime, let ζ be a primitive p-th root of unity, and let

$$T_p = \sum_{r=0}^{p-1} \frac{\zeta^{r^2}}{(-\zeta;\zeta)_r^2} \in \mathbb{Z}[\zeta]$$

be the cyclotomic value attached to Ramanujan's third-order mock theta function $f(q)$ by the odd-order radial-limit theorem of [1], where $\lim_{r\to 1^-} f(r\zeta) = \frac{4}{3}T_p$. Writing $T_p = \sum_{m=0}^{p-2} c_m \zeta^m$ in the power basis $\{1, \zeta, \ldots, \zeta^{p-2}\}$, the constant coordinate $c_0(T_p)$ was conjectured in [1] to equal 3 for every prime $p \geq 7$ (verified there exhaustively to $p \leq 10^4$), and identified as the one arithmetic invariant of the "mock Gauss sum" T_p that is provably invisible to every symmetric function of its Galois conjugates.

We prove the conjecture. The proof is a finite specialization of the Rogers–Fine identity. Its output is the exact weighted quadratic formula

$$\boxed{T_p = \frac{1}{p}\sum_{n=0}^{p-1}\left(\frac{3p-1}{2} - 3n\right)\zeta^{-n(3n+1)/2}}$$

valid for every prime $p \geq 7$, from which $c_0(T_p) = 3$ follows because only four quadratic residues contribute to the two relevant redundant coordinates. The argument uses no asymptotics, no Eichler integral, and no unverified analytic input; the Fine collapse and the Rogers–Fine identity are established as *finite polynomial q-difference identities*, and the "ℓ'Hôpital" step of the classical derivation is replaced by a single formal differentiation. The complete argument has been formalized in Lean 4 over Mathlib; the formalization is machine-checked, contains no `sorry`, `admit`, or added axioms, and additionally certifies that every power-basis coefficient of T_p lies in $\{0, 1, 2, 3\}$ and that $\mathrm{Tr}_{\mathbb{Q}(\zeta)/\mathbb{Q}}(T_p)$ equals $2p - 1$ or $p - 1$ according to p mod 6.

Contents

1 Introduction

Ramanujan's third-order mock theta function is

$$f(q) \;=\; \sum_{n\ge 0}\frac{q^{n^2}}{(-q;q)_n^2}, \qquad (a;q)_n := \prod_{j=0}^{n-1}(1-aq^j).$$

At a primitive p-th root of unity ζ the series is singular, but its radial limit exists and is governed by a finite cyclotomic element. Writing $(-\zeta;\zeta)_r = \prod_{j=1}^{r}(1+\zeta^j)$, the odd-order radial-limit theorem of [1] gives, for every odd prime p,

$$\lim_{r\to 1^-} f(r\zeta) \;=\; \tfrac{4}{3}\,T_p, \qquad T_p := \sum_{r=0}^{p-1}\frac{\zeta^{r^2}}{(-\zeta;\zeta)_r^2} \;\in\; \mathbb{Z}[\zeta]. \tag{1}$$

The element T_p lies in the ring of integers of $\mathbb{Q}(\zeta)$ and, in the power basis $\{1,\zeta,\ldots,\zeta^{p-2}\}$, has a distinguished constant coordinate

$$c_0(T_p)\in\mathbb{Z}, \qquad \text{where } T_p = \sum_{m=0}^{p-2} c_m\zeta^m.$$

The problem

The paper [1] carried out an extensive study of $c_0(T_p)$, established a chain of unconditional reductions, and framed T_p as a *mock Gauss sum*: modulo a prime above 3 it collapses to a shifted quadratic Gauss sum, and conjecturally lifts in characteristic 0 to a Gauss-sum part plus a 0/1 "mock" correction. Two features made the constant coordinate resistant to standard tools.

1. *Galois non-invariance.* It was proved in [1] that $c_0(T_p)$ is not a symmetric function of the conjugates of T_p: it is determined neither by the trace, the norm, the characteristic polynomial, nor any class L-value. Concretely, for the automorphism $\sigma_2 : \zeta \mapsto \zeta^2$ one has $c_0(\sigma_2 T_p) = c_0(T_p) - c_{(p-1)/2}(T_p)$, so any presentation of c_0 through symmetric data must fail.

2. *Moment obstruction.* Standard moment/analytic estimates bound c_0 only to size $O(\sqrt{p})$, far from the constant value 3.

The conjecture $c_0(T_p) = 3$ was verified computationally to $p \le 10^4$ and left open, with the residual difficulty isolated as a single analytic computation (an odd-order additive-twisted Eichler-integral evaluation).

Result

We resolve the conjecture by producing the missing exact identity directly, bypassing the analytic route.

Theorem 1.1 (Main theorem). *For every prime $p \geq 7$ and every primitive p-th root of unity ζ,*

$$c_0(T_p) = 3.$$

The engine is a finite weighted quadratic identity, whose statement should be compared with a Gauss sum $\sum_n \zeta^{-n(3n+1)/2}$ carrying a linear weight.

Theorem 1.2 (Weighted Rogers–Fine formula). *For every prime $p \geq 7$,*

$$T_p \;=\; \frac{1}{p}\sum_{n=0}^{p-1}\Big(\tfrac{3p-1}{2} - 3n\Big)\,\zeta^{-n(3n+1)/2}. \tag{2}$$

The quadratic exponent $A_n := n(3n+1)/2$ is Euler's pentagonal exponent. Passing to the redundant spanning set $\{1, \zeta, \ldots, \zeta^{p-1}\}$, the coefficient of ζ^0 receives contributions from the residues n with $A_n \equiv 0 \pmod{p}$, and the coefficient of $\zeta^{-1} = \zeta^{p-1}$ from those with $A_n \equiv 1$. Because $p \neq 3$, each congruence has exactly two solutions, and the linear weight collapses their difference to $3(p-1) + 3 = 3p$; dividing by p gives Theorem 1.1.

Method and rigor

Equation (2) is a specialization of the classical Rogers–Fine identity. At a root of unity the naïve derivation involves radial limits, a $0/0$ indeterminacy resolved by ℓ'Hôpital, and a removable singularity in one summand; each is a genuine analytic subtlety. We avoid all three. The two q-series identities we need—Fine's collapse and the Rogers–Fine identity—are recast as *finite polynomial identities* in $K[X]$, where K is any characteristic-zero field containing a primitive p-th root of unity, and are proved by first-order polynomial q-difference equations together with a uniqueness lemma (Sections 2–4). The weighted formula (2) then arises by differentiating a single polynomial identity once and evaluating at $X = 1$ (Section 5); the removable singularity disappears because polynomial differentiation never divides. The constant-coordinate extraction (Section 6) is elementary linear algebra over $\mathbb{Q}$.

Every step has been formalized in Lean 4 over Mathlib (Section 8). The formalization proves Theorem 1.1 for all primes $p \geq 7$ in every characteristic-zero cyclotomic extension of $\mathbb{Q}$; it has no `sorry`, `admit`, or additional axioms, and its principal-theorem axiom audit reports only Lean's three standard axioms.

Relation to the mock Gauss-sum programme

Formula (2) is precisely the exact "theta-like" object that the analytic programme of [1] sought: the mock correction is realized concretely as the linear weight $\frac{3p-1}{2} - 3n$ attached to a pentagonal Gauss sum, rather than as an additive-twisted Eichler integral. The Galois non-invariance obstruction of [1] is respected—the proof genuinely uses non-symmetric data, namely the ordering of the exponents A_n modulo p—but it does so through a finite quadratic sum rather than a transcendental period. We discuss this, and correct one auxiliary claim of [1], in Section 9.

Notation. Throughout, $p \geq 7$ is a prime and ζ a primitive p-th root of unity in a field of characteristic 0. We write $u := \zeta^{-1}$, which is again a primitive p-th root of unity, and set

$$P_n \;:=\; (-u;u)_n \;=\; \prod_{j=1}^{n}\bigl(1+u^j\bigr), \qquad P_0 = 1.$$

For $f \in K[X]$ we write $f(uX)$ for the image of f under $X \mapsto uX$.

2 Cyclotomic product identities

We first record the two elementary product identities on which everything rests. Let K be a field of characteristic 0 and $u \in K$ a primitive p-th root of unity with p odd.

Lemma 2.1. *For every $j \geq 0$ we have $1 + u^j \neq 0$.*

Proof. If $1 + u^j = 0$ then $u^j = -1$, hence $1 = (u^j)^p = (-1)^p = -1$ because p is odd, contradicting $\operatorname{char} K = 0$. □

Lemma 2.2 (Full cycle product)**.** *In $K[X]$,*

$$\prod_{j=0}^{p-1}\bigl(1+u^jX\bigr) \;=\; 1+X^p.$$

Proof. Since u is a primitive p-th root of unity and p is odd, the elements $-u^0, \ldots, -u^{p-1}$ are exactly the roots of $X^p + 1$, so

$$X^p + 1 = \prod_{j=0}^{p-1}\bigl(X+u^j\bigr).$$

Applying the degree-p reversal $g(X) \mapsto X^{\deg g} g(1/X)$ to both sides—which sends $X^p + 1$ to $1 + X^p$ and each factor $X + u^j$ to $1 + u^j X$—gives the claim. □

Lemma 2.3 (Cyclotomic defect)**.** $\displaystyle\prod_{j=1}^{p-1}\bigl(1+u^j\bigr) = 1$*; equivalently $P_{p-1} = 1$. Moreover $P_p = 2$.*

Proof. Evaluate Lemma 2.2 at $X = 1$: $\prod_{j=0}^{p-1}(1+u^j) = 1+1 = 2$. The $j = 0$ factor equals $1+u^0 = 2$, and $2 \neq 0$ in K, so cancelling it gives $\prod_{j=1}^{p-1}(1+u^j) = 1$. Finally $P_p = P_{p-1}\,(1+u^p) = 1 \cdot 2 = 2$ since $u^p = 1$. □

In particular $P_n \neq 0$ for all n (a product of nonzero factors), and the "defect" $P_{p-1} = 1$ makes the sequence P_n periodic up to scale: $P_{n+p} = 2P_n$.

Remark 2.4. Lemma 2.3 holds for every *odd* order, prime or composite. This corrects a claim in [1] to the effect that $\prod_{j=1}^{k-1}(1+\zeta^j)$ fails to equal 1 for composite k and could serve as a primality test; the product equals 1 for all odd $k > 1$. See Section 9.

3 The finite Fine collapse

Fine's classical identity for the mock theta function is

$$f(q) = 1 + \sum_{n\geq 1} \frac{(-1)^{n-1}q^n}{(-q;q)_n}$$

[2, Ch. 3], [4]. Specialized through (1), it collapses T_p to a linear Pochhammer sum. We prove the finite form directly as a polynomial identity, with no appeal to convergence. Throughout this section write

$$Q_n := (-\zeta;\zeta)_n^{-1} = \prod_{j=1}^{n} \left(1+\zeta^j\right)^{-1}, \qquad Q_0 = 1,$$

and define in $K[X]$ the generating polynomials

$$\mathcal{L}(X) := \sum_{n=0}^{p-1} (-1)^n Q_n X^n,$$

$$\mathcal{R}(X) := \sum_{n=0}^{p-1} Q_n \zeta^{n^2} X^n \prod_{k=n+1}^{p-1} \left(1+\zeta^k X\right).$$

Thus $\mathcal{R}$ is the finite "other side" of Fine's identity, written in base ζ; in particular, its summands carry no factor $(-1)^n$.

Lemma 3.1 (q-difference equations)**.** *Both $\mathcal{L}$ and $\mathcal{R}$ satisfy*

$$(1+X)F(X) + F(\zeta X) = 2 + X^p. \tag{3}$$

Proof. Let

$$\lambda_n(X) := (-1)^n Q_n X^n.$$

For $0 \leq n \leq p-2$,

$$\lambda_{n+1}(X) + \lambda_{n+1}(\zeta X) = -X\lambda_n(X),$$

because $Q_{n+1}(1+\zeta^{n+1}) = Q_n$. Consequently all interior terms in $(1+X)\mathcal{L}(X) + \mathcal{L}(\zeta X)$ cancel. The constant boundary contributes $2Q_0 = 2$, while the upper boundary contributes X^p, since $p-1$ is even and

$$Q_{p-1} = \left(\prod_{j=1}^{p-1}(1+\zeta^j)\right)^{-1} = 1$$

by Lemma 2.3. Hence

$$(1+X)\mathcal{L}(X) + \mathcal{L}(\zeta X) = 2 + X^p.$$

For the right side, let

$$\rho_n(X) := Q_n \zeta^{n^2} X^n \prod_{k=n+1}^{p-1} (1+\zeta^k X)$$

and introduce the certificate

$$\gamma_n(X) := -(1+X)(1+\zeta^n)\rho_n(X).$$

For $0 \le n \le p-2$, direct substitution gives

$$(1+X)\rho_n(X) + \rho_n(\zeta X) = \gamma_{n+1}(X) - \gamma_n(X). \tag{4}$$

Indeed, this is obtained from

$$Q_{n+1}(1+\zeta^{n+1}) = Q_n, \qquad (n+1)^2 = n^2 + 2n + 1,$$

together with

$$\prod_{k=n+1}^{p-1}(1+\zeta^k X) = (1+\zeta^{n+1}X)\prod_{k=n+2}^{p-1}(1+\zeta^k X)$$

and the corresponding shift under $X \mapsto \zeta X$.

The last summand satisfies the separate boundary identity

$$(1+X)\rho_{p-1}(X) + \rho_{p-1}(\zeta X) = -X^p - \gamma_{p-1}(X). \tag{5}$$

Here one uses $Q_{p-1} = 1$, $\zeta^{(p-1)^2} = \zeta$, and $\zeta^{p-1}\zeta = 1$. Summing (4) and adding (5) telescopes to

$$(1+X)\mathcal{R}(X) + \mathcal{R}(\zeta X) = -\gamma_0(X) - X^p.$$

Finally,

$$\gamma_0(X) = -2(1+X)\prod_{k=1}^{p-1}(1+\zeta^k X) = -2(1+X^p)$$

by Lemma 2.2. Therefore

$$(1+X)\mathcal{R}(X) + \mathcal{R}(\zeta X) = 2 + X^p.$$

□

Lemma 3.2 (Uniqueness)**.** *If $H \in K[X]$ satisfies*

$$(1+X)H(X) + H(\zeta X) = 0,$$

then $H = 0$.

Proof. Write $H(X) = \sum_{n\ge 0} h_n X^n$. Comparing constant coefficients gives $2h_0 = 0$, hence $h_0 = 0$. Comparing coefficients of X^{n+1} gives

$$(1+\zeta^{n+1})h_{n+1} + h_n = 0.$$

By induction $h_n = 0$, and Lemma 2.1 gives $1+\zeta^{n+1} \neq 0$, so $h_{n+1} = 0$. Thus every coefficient of H vanishes. □

Proposition 3.3 (Finite Fine identity)**.** *In $K[X]$,*

$$\mathcal{L}(X) = \mathcal{R}(X).$$

Consequently,

$$T_p = \sum_{n=0}^{p-1}(-1)^n Q_n.$$

Proof. By Lemma 3.1, the difference $H = \mathcal{L} - \mathcal{R}$ satisfies

$$(1+X)H(X) + H(\zeta X) = 0.$$

Lemma 3.2 therefore gives $H = 0$.

Evaluate at $X = 1$. Clearly,

$$\mathcal{L}(1) = \sum_{n=0}^{p-1} (-1)^n Q_n.$$

For the other side, Lemma 2.3 gives

$$\prod_{k=n+1}^{p-1} (1+\zeta^k) = \frac{\prod_{k=1}^{p-1}(1+\zeta^k)}{\prod_{k=1}^{n}(1+\zeta^k)} = Q_n.$$

Hence the n-th summand of $\mathcal{R}(1)$ equals

$$Q_n \zeta^{n^2} Q_n = \frac{\zeta^{n^2}}{(-\zeta;\zeta)_n^2}.$$

Thus $\mathcal{R}(1) = T_p$, proving the collapse. $\square$

Reindexing $n \mapsto p-1-n$ in Proposition 3.3 gives the form used below. Since $p-1$ is even,

$$(-1)^{p-1-n} = (-1)^n,$$

and Lemma 2.3 gives

$$\begin{aligned} Q_{p-1-n} &= \prod_{k=1}^{p-1-n} (1+\zeta^k)^{-1} \\ &= \prod_{k=p-n}^{p-1} (1+\zeta^k) \\ &= \prod_{j=1}^{n} (1+\zeta^{-j}). \end{aligned}$$

Proposition 3.4 (Reverse-Fine collapse)**.** *For every odd prime p,*

$$T_p = \sum_{n=0}^{p-1} (-1)^n P_n, \qquad P_n = \prod_{j=1}^{n} (1+\zeta^{-j}). \tag{6}$$

4 The finite Rogers–Fine identity

The Rogers–Fine identity [2, Ch. 14], [3] is

$$\sum_{n\ge 0} \frac{(aq;q)_n}{(bq;q)_n} t^n = \sum_{n\ge 0} \frac{(aq;q)_n (atq/b;q)_n \, b^n t^n q^{n^2} \left(1 - atq^{2n+1}\right)}{(bq;q)_n \, (t;q)_{n+1}}.$$

Taking the confluent limit $b \to 0$, then $a = -1$, $q = u$, $t = -X$, gives (after clearing denominators) a *polynomial* identity that we now state and prove directly. Define the reverse-Fine generating polynomial

$$L(X) := \sum_{n=0}^{p-1} (-1)^n P_n \, X^n,$$

so that $L(1) = T_p$ by Proposition 3.4, and the denominator-cleared right-hand side

$$R(X) := \sum_{n=0}^{p-1} P_n \, u^{A_n} \, X^{2n} \big(1 - u^{2n+1} X\big) \prod_{k=n+1}^{p-1} \big(1 + u^k X\big), \qquad A_n = \tfrac{n(3n+1)}{2}.$$

Lemma 4.1 (q-difference equations)**.** *In* $K[X]$,

$$(1 + X)\, L(X) + uX\, L(uX) = 1 + 2X^p, \tag{7}$$

$$(1 + X)\, R(X) + uX\, R(uX) = 1 + X^p - 2X^{2p}. \tag{8}$$

Proof. For (7), put

$$\ell_n(X) := (-1)^n P_n X^n.$$

The recurrence $P_{n+1} = P_n(1 + u^{n+1})$ gives

$$X\ell_n(X) + uX\ell_n(uX) = -\ell_{n+1}(X).$$

Summing this relation for $0 \le n \le p-1$ leaves the two boundary terms $\ell_0 = 1$ and $-\ell_p = 2X^p$, since p is odd and $P_p = 2$. This proves (7).

For (8), write

$$\Pi_n(X) := \prod_{k=n+1}^{p-1} (1 + u^k X)$$

and define the summand and certificate by

$$\varrho_n(X) := P_n u^{A_n} X^{2n} (1 - u^{2n+1} X) \Pi_n(X),$$
$$g_n(X) := P_n u^{A_n} X^{2n} (1 + X) \Pi_n(X).$$

Thus $R(X) = \sum_{n=0}^{p-1} \varrho_n(X)$. We claim that, for $0 \le n \le p-2$,

$$(1 + X)\varrho_n(X) + uX\,\varrho_n(uX) = g_n(X) - g_{n+1}(X). \tag{9}$$

Indeed,

$$\Pi_n(X) = (1 + u^{n+1} X)\Pi_{n+1}(X), \qquad \Pi_n(uX) = (1 + X)\Pi_{n+1}(X),$$

where the last factor in the shifted product is $1 + u^p X = 1 + X$. Factoring the left side of (9) gives

$$\begin{aligned}
&(1 + X)\varrho_n(X) + uX\,\varrho_n(uX) \\
&\quad = P_n u^{A_n} X^{2n} (1 + X) \Pi_{n+1}(X) \\
&\qquad \times \Big[(1 - u^{2n+1} X)(1 + u^{n+1} X) + u^{2n+1} X (1 - u^{2n+2} X) \Big] \\
&\quad = P_n u^{A_n} X^{2n} (1 + X) \Pi_{n+1}(X) \\
&\qquad \times \Big[1 + u^{n+1} X - (1 + u^{n+1}) u^{3n+2} X^2 \Big].
\end{aligned}$$

The first two terms in the last bracket give $g_n(X)$. Using

$$P_{n+1} = P_n(1 + u^{n+1}), \qquad A_{n+1} = A_n + 3n + 2,$$

the remaining term is $-g_{n+1}(X)$, proving the certificate identity.

The final summand is a separate boundary calculation. Set $n = p-1$. Then $P_n = 1$ and $\Pi_n = 1$. Moreover

$$u^{A_p} = 1, \qquad A_p = A_n + 3n + 2, \qquad n + 1 = p,$$

so $u^p = 1$ implies

$$u^{A_n+2n+1} = 1, \qquad u^{A_n+4n+3} = 1.$$

Consequently

$$\begin{aligned}
&(1+X)\varrho_{p-1}(X) + uX\,\varrho_{p-1}(uX) \\
&= X^{2n}\Big[u^{A_n}(1+X) - \big(u^{A_n+2n+1} + u^{A_n+4n+3}\big)X^2\Big] \\
&= g_{p-1}(X) - 2X^{2p}.
\end{aligned}$$

At the initial boundary,

$$g_0(X) = (1+X)\prod_{k=1}^{p-1}(1+u^kX) = 1 + X^p$$

by Lemma 2.2. Summing (9) for $0 \le n \le p-2$ and adjoining the final boundary therefore telescopes to

$$(1+X)R(X) + uXR(uX) = g_0(X) - 2X^{2p} = 1 + X^p - 2X^{2p},$$

which is (8). □

Lemma 4.2 (Uniqueness)**.** *If $H \in K[X]$ satisfies $(1+X)H(X) + uX\,H(uX) = 0$, then $H = 0$.*

Proof. Coefficient of X^0: $H_0 = 0$. Coefficient of X^{n+1}: $H_{n+1} + H_n + u \cdot u^n H_n = 0$, i.e. $H_{n+1} = -(1+u^{n+1})H_n$; by induction $H_n = 0$ for all n. □

Proposition 4.3 (Finite Rogers–Fine identity)**.** *In $K[X]$,*

$$\big(1 - X^p\big)\,L(X) \;=\; R(X). \tag{10}$$

Proof. Since $u^p = 1$, the substitution $X \mapsto uX$ fixes $1 - X^p$. Hence, using (7),

$$(1+X)\big[(1-X^p)L\big] + uX\big[(1-X^p)L\big](uX) = (1-X^p)\big[(1+X)L + uX\,L(uX)\big] = (1-X^p)(1+2X^p),$$

which equals $1 + X^p - 2X^{2p}$. By (8), the polynomial $H = (1-X^p)L - R$ satisfies $(1+X)H + uX\,H(uX) = 0$, so $H = 0$ by Lemma 4.2. □

5 The weighted quadratic formula

We now differentiate (10) once and evaluate at $X = 1$. This replaces the $0/0$ limit and ℓ'Hôpital step of the analytic derivation; because we differentiate a polynomial identity, no denominator ever vanishes and the summand at $n = (p-1)/2$ (where $1 - u^{2n+1} = 0$) needs no special treatment.

Let ∂ denote the formal derivative on $K[X]$. Differentiating (10) and evaluating at $X = 1$: the left side gives

$$\partial\big[(1-X^p)L\big]\big|_{X=1} = -p\,L(1) + 0 = -p\,T_p,$$

since $(1-X^p)|_{X=1} = 0$ and $\partial(1-X^p)|_{X=1} = -p$. The right side is $\partial R|_{X=1} = \sum_{n=0}^{p-1} \partial\varrho_n|_{X=1}$.

Lemma 5.1 (Termwise derivative)**.** *For $0 \le n \le p-1$, define the tail logarithmic derivative*

$$T_{p,n} := \sum_{k=n+1}^{p-1} \frac{u^k}{1+u^k}.$$

Then

$$\partial\varrho_n(1) = u^{A_n}\left[2n\big(1-u^{2n+1}\big) - u^{2n+1} + \big(1-u^{2n+1}\big)T_{p,n}\right].$$

Proof. Write

$$\varrho_n(X) = P_n u^{A_n} X^{2n}(1-u^{2n+1}X)\Pi_n(X), \qquad \Pi_n(X) := \prod_{k=n+1}^{p-1}(1+u^kX).$$

The product rule gives

$$\partial\Pi_n(1) = \Pi_n(1)T_{p,n}.$$

Moreover,

$$\Pi_n(1) = P_n^{-1}$$

by Lemma 2.3. Thus the factors P_n and $\Pi_n(1)$ cancel. The derivatives of X^{2n}, $1-u^{2n+1}X$, and $\Pi_n(X)$ contribute, respectively,

$$2n(1-u^{2n+1}), \qquad -u^{2n+1}, \qquad (1-u^{2n+1})T_{p,n}.$$

Adding them proves the formula. No division by $1-u^{2n+1}$ occurs, so the index $n=(p-1)/2$ requires no separate treatment. ☐

The next two identities are the finite reflection symmetries that make the sum collapse. They are the exact analogues of the classical facts $S_{p-1} = p/2$ and $S_n - S_{p-1-n} = n - \frac{p-1}{2}$.

Lemma 5.2 (Reflection symmetries)**.** *For $0 \le n \le p-1$:*

(i) $\dfrac{u^j}{1+u^j} + \dfrac{u^{-j}}{1+u^{-j}} = 1$, *hence* $\displaystyle\sum_{j=0}^{p-1}\frac{u^j}{1+u^j} = \frac{p}{2}$*;*

(ii) $u^{A_n+2n+1} = u^{A_{p-1-n}}$, *equivalently* $A_n + 2n + 1 \equiv A_{p-1-n} \pmod p$*;*

(iii) $T_{p,n} - T_{p,\,p-1-n} = \dfrac{p-1}{2} - n.$

Proof. (i) Multiply numerator and denominator of the second term by u^j to get $1/(1+u^j)$, and add. Pairing $j \leftrightarrow p-j$ over $1 \le j \le p-1$ (each pair summing to 1) and adding the $j=0$ term $\frac{1}{2}$ gives $\frac{p-1}{2} + \frac{1}{2} = \frac{p}{2}$. (ii) Both sides double to $2A_n + 2(2n+1) = n(3n+1) + 2(2n+1) = 3n^2 + 5n + 2 = (n+1)(3n+2)$ and $2A_{p-1-n} = (p-1-n)(3(p-1-n)+1)$; modulo p these agree because $(p-1-n) \equiv -(1+n)$ and $3(p-1-n)+1 \equiv -(3n+2)$, whose product is $(n+1)(3n+2)$. As $\gcd(2,p)=1$ the congruence descends. Raising u to both sides gives the exponent identity. (iii) Put

$$f_j := \frac{u^j}{1+u^j}, \qquad m := p-1-n.$$

Reflect the finite range $n+1 \le r \le p-1$ by writing $r = p-j$. As j then runs through $1, \dots, m$, part (i) gives

$$T_{p,n} = \sum_{r=n+1}^{p-1} f_r = \sum_{j=1}^{m} f_{p-j} = \sum_{j=1}^{m} (1 - f_j)$$
$$= m - \sum_{j=1}^{m} f_j.$$

On the other hand, split the full finite cycle at m:

$$\frac{p}{2} = \sum_{j=0}^{p-1} f_j = f_0 + \sum_{j=1}^{m} f_j + \sum_{j=m+1}^{p-1} f_j = \frac{1}{2} + \sum_{j=1}^{m} f_j + T_{p,m}.$$

Thus $\sum_{j=1}^{m} f_j + T_{p,m} = (p-1)/2$, and hence

$$T_{p,n} - T_{p,m} = m - \frac{p-1}{2} = \frac{p-1}{2} - n.$$

Since $m = p-1-n$, this is precisely the asserted identity. □

Theorem 5.3 (Weighted formula; = Theorem 1.2)**.**

$$T_p = \frac{1}{p} \sum_{n=0}^{p-1} \left(\frac{3p-1}{2} - 3n \right) u^{A_n} = \frac{1}{p} \sum_{n=0}^{p-1} \left(\frac{3p-1}{2} - 3n \right) \zeta^{-A_n}.$$

Proof. Summing Lemma 5.1 and comparing with

$$-pT_p = \partial R|_{X=1},$$

we obtain

$$-pT_p = \sum_{n=0}^{p-1} u^{A_n} \left[2n(1 - u^{2n+1}) - u^{2n+1} + (1 - u^{2n+1}) T_{p,n} \right]. \tag{11}$$

We simplify the polynomial part and the $T_{p,n}$-part separately.

For the polynomial part,

$$\sum_{n=0}^{p-1} u^{A_n} \left[2n - (2n+1) u^{2n+1} \right]$$
$$= \sum_{n=0}^{p-1} 2n\, u^{A_n} - \sum_{n=0}^{p-1} (2n+1) u^{A_n + 2n+1}.$$

By Lemma 5.2(ii),

$$u^{A_n + 2n+1} = u^{A_{p-1-n}}.$$

Reindexing the second sum by $m = p-1-n$ turns it into

$$\sum_{m=0}^{p-1} (2p - 1 - 2m) u^{A_m}.$$

Therefore the polynomial part equals

$$\sum_{n=0}^{p-1}(4n-2p+1)u^{A_n}. \tag{7.1}$$

For the tail part,

$$\begin{aligned}&\sum_{n=0}^{p-1} u^{A_n}(1-u^{2n+1})T_{p,n}\\ &\qquad = \sum_{n=0}^{p-1} u^{A_n}T_{p,n} - \sum_{n=0}^{p-1} u^{A_n+2n+1}T_{p,n}.\end{aligned}$$

Using Lemma 5.2(ii) and reindexing the second sum gives

$$\sum_{n=0}^{p-1} u^{A_n}\big(T_{p,n}-T_{p,p-1-n}\big).$$

Lemma 5.2(iii) therefore yields

$$\sum_{n=0}^{p-1}\left(\frac{p-1}{2}-n\right)u^{A_n}. \tag{7.2}$$

Adding (7.1) and (7.2),

$$\begin{aligned}-pT_p &= \sum_{n=0}^{p-1}\left(4n-2p+1+\frac{p-1}{2}-n\right)u^{A_n}\\ &= \sum_{n=0}^{p-1}\left(3n-\frac{3p-1}{2}\right)u^{A_n}\\ &= -\sum_{n=0}^{p-1}\left(\frac{3p-1}{2}-3n\right)u^{A_n}.\end{aligned}$$

Dividing by $-p$ proves the first equality. The second follows from $u=\zeta^{-1}$. □

6 Extraction of the constant coordinate

We convert (2) from the redundant spanning set $\{1,\zeta,\dots,\zeta^{p-1}\}$ to the power basis $\{1,\zeta,\dots,\zeta^{p-2}\}$ and read off c_0.

Lemma 6.1 (Redundant-to-power basis)**.** *For any coefficients $C_0,\dots,C_{p-1}\in\mathbb{Q}$,*

$$c_0\left(\sum_{m=0}^{p-1} C_m\,\zeta^m\right) = C_0 - C_{p-1}.$$

Proof. For $0\le m\le p-2$, ζ^m is a basis vector, contributing C_m to its own coordinate and 0 to c_0 unless $m=0$. The single redundant power is $\zeta^{p-1}=-(1+\zeta+\cdots+\zeta^{p-2})$, whose constant coordinate is -1. Hence $c_0 = C_0\cdot 1 + C_{p-1}\cdot(-1)$. □

By (2), $T_p = \sum_{m=0}^{p-1} C_m \zeta^m$, where the general redundant coefficient is

$$C_m = \frac{1}{p} \sum_{\substack{0 \le n < p \\ -A_n \equiv m \pmod p}} w_n = \frac{1}{p} \sum_{\substack{0 \le n < p \\ A_n \equiv -m \pmod p}} w_n, \qquad w_n = \frac{3p-1}{2} - 3n.$$

In particular, C_0 is supported on $A_n \equiv 0$, while C_{p-1} is supported on $A_n \equiv 1$, because $-A_n \equiv p-1$ is equivalent to $A_n \equiv 1$. Thus, tracking ζ^0 and $\zeta^{-1} = \zeta^{p-1}$,

$$c_0(T_p) = \frac{1}{p}\Big(\sum_{n\colon A_n \equiv 0} w_n - \sum_{n\colon A_n \equiv 1} w_n \Big),$$

the two sums over $0 \le n \le p-1$.

Lemma 6.2 (Four quadratic roots)**.** *Let $p \ge 7$ and let $a \in \{0, \dots, p-1\}$ be the canonical representative of $-3^{-1} \pmod p$. Then, for $0 \le n \le p-1$,*

$$A_n \equiv 0 \pmod p \iff n \in \{0,\ a\}, \qquad A_n \equiv 1 \pmod p \iff n \in \{p-1,\ a+1\},$$

and the four residues 0, a, $p-1$, $a+1$ are pairwise distinct.

Proof. Since $\gcd(2,p) = 1$, the congruence $A_n \equiv 0$ is equivalent to $n(3n+1) \equiv 0$. Its solutions are $n \equiv 0$ and $n \equiv -3^{-1} = a$. Similarly,

$$A_n \equiv 1 \iff 3n^2 + n - 2 \equiv 0 \iff (3n-2)(n+1) \equiv 0.$$

The two solutions are $n \equiv -1 = p-1$ and $n \equiv 2 \cdot 3^{-1}$. Because

$$a + 1 \equiv -3^{-1} + 1 = (-1+3)3^{-1} = 2 \cdot 3^{-1} \pmod p,$$

the latter solution is represented by $a+1$.

It remains to check that these are four distinct representatives. The relation $3a \equiv -1$ shows $a \ne 0$. If $a = p-1$, then $-3 \equiv -1 \pmod p$, so $p \mid 2$, impossible for $p \ge 7$. Hence $a \ne p-1$, and since $0 \le a < p$, we have $a+1 < p$; thus $a+1$ is the actual representative in $\{0, \dots, p-1\}$. Now $a+1$ differs from a as an integer and is nonzero. Finally, if $a+1 = p-1$, then $2 \cdot 3^{-1} \equiv -1 \pmod p$; multiplying by 3 gives $2 \equiv -3 \pmod p$, so $p \mid 5$, again impossible for $p \ge 7$. Together with $0 \ne p-1$, this proves pairwise distinctness. □

Proof of Theorem 1.1. By Lemma 6.2 the two sums are over $\{0, a\}$ and $\{p-1, a+1\}$, so

$$c_0(T_p) = \frac{1}{p}\big[(w_0 + w_a) - (w_{p-1} + w_{a+1})\big] = \frac{1}{p}\big[(w_0 - w_{p-1}) + (w_a - w_{a+1})\big].$$

The weight $w_n = \frac{3p-1}{2} - 3n$ is affine with slope -3, so $w_0 - w_{p-1} = 3(p-1)$ and $w_a - w_{a+1} = 3$. Therefore

$$c_0(T_p) = \frac{3(p-1)+3}{p} = \frac{3p}{p} = 3.$$

□

7 Corollaries

The same coordinate bookkeeping yields several global consequences for the coefficient vector. The following are proved in the same framework and formalized (Section 8); we state them for completeness. Exact coefficient multiplicities are not yet formalized.

Proposition 7.1 (Coefficient classification). *For every prime $p \geq 7$, each power-basis coefficient $c_m(T_p)$ lies in $\{0, 1, 2, 3\}$.*

Before stating the remaining formulas, define the sum of the power-basis coefficients by

$$T_p(1) := \sum_{m=0}^{p-2} c_m(T_p).$$

Proposition 7.2 (Coefficient sum). *For every prime $p \geq 7$,*

$$T_p(1) = \begin{cases} p+1, & p \equiv 1 \pmod 6, \\ 2p+1, & p \equiv 5 \pmod 6. \end{cases}$$

Proposition 7.3 (Trace). *For every prime $p \geq 7$,*

$$\mathrm{Tr}_{\mathbb{Q}(\zeta)/\mathbb{Q}}(T_p) = \begin{cases} 2p-1, & p \equiv 1 \pmod 6, \\ p-1, & p \equiv 5 \pmod 6. \end{cases}$$

Propositions 7.2 and 7.3 satisfy the character-free reformulation $T_p(1) + \mathrm{Tr}(T_p) = 3p$ of [1]: indeed $p\, c_0(T_p) = T_p(1) + \mathrm{Tr}(T_p)$ from Lemma 6.1 summed over all coordinates, and $c_0 = 3$ gives the result.

8 The Lean 4 formalization

The entire argument above has been formalized in Lean 4 over Mathlib (toolchain `v4.31.0`). The development is self-contained apart from Mathlib and reproduces, as ordinary finite algebra, every identity of Sections 2–6. The main results are:

- `MockTheta.fullCycleProd`, `cyclotomicDefect` — Lemmas 2.2–2.3;
- `MockTheta.finiteFine`, `fineCollapse`, `reverseFine` — Proposition 3.4, via the q-difference equations (3) and uniqueness (Lemma 3.2);
- `MockTheta.finiteRogersFine` — Proposition 4.3, the denominator-cleared identity (10);
- `MockTheta.weighted_formula`. This proves Theorem 1.2 by formal differentiation and evaluation at $X = 1$;
- `MockTheta.c0_T_eq_three` (public alias `constantCoeff_T_eq_three`) — Theorem 1.1, valid for every prime $p \geq 7$ in every characteristic-zero cyclotomic extension containing a primitive p-th root;
- `MockTheta.TAtOne_of_eq_six_mul_add_one/five` — Proposition 7.2, the two coefficient-sum formulas;

- `MockTheta.cCoeff_T_mem_four` — Proposition 7.1, the universal coefficient classification;
- `MockTheta.trace_T_of_eq_six_mul_add_one/five` — Proposition 7.3, using Mathlib's algebra trace.

The development does not yet prove exact coefficient multiplicities, the characteristic-zero mock Gauss-sum lift, or the subset-count corollaries; these remain downstream results and are not used above.

The constant coordinate is defined honestly as the coefficient of 1 in Mathlib's cyclotomic power basis,

$$\texttt{c0}\ (h\zeta)\ x \ = \ (h\zeta\texttt{.powerBasis}\ \mathbb{Q})\texttt{.basis.repr}\ x\ \langle 0, \dots \rangle,$$

so `c0_T_eq_three` literally asserts $c_0(T_p) = 3$. The proof of $c_0(\zeta^{p-1}) = -1$ (Lemma 6.1) is derived from $\zeta^{p-1} = -(1 + \cdots + \zeta^{p-2})$.

The development contains no `sorry`, `admit`, added `axiom`, or proof by `native_decide`. An `#print axioms` audit of the principal theorem reports only Lean's three foundational axioms `propext`, `Classical.choice`, and `Quot.sound`. Typechecked regression examples at $p = 7, 11, 13$ are included. The finite q-difference method means the formalization uses *no* analytic convergence, radial limits, or ℓ'Hôpital; the removable singularity at $n = (p-1)/2$ noted after Lemma 5.1 requires no special case because differentiation is formal.

9 Remarks and relation to the mock Gauss-sum programme

The mock correction, made explicit. The programme of [1] predicted that T_p lifts in characteristic 0 to a Gauss-sum part plus a 0/1 mock indicator, with the constant coordinate of the Gauss-sum part vanishing (an exact consequence of $1 + 24 = 25 = 5^2$) so that $c_0(T_p)$ equals the sole surviving indicator value, conjecturally 3. Theorem 1.2 realizes this picture concretely: T_p *is* a pentagonal Gauss sum $\sum_n \zeta^{-A_n}$ dressed by the affine weight $\frac{3p-1}{2} - 3n$, and the constant coordinate is exactly the weighted count of the four quadratic residues in Lemma 6.2. No Eichler integral is needed; the "one remaining computation" of [1] is discharged by the finite identity (10).

Consistency with the obstruction. The Galois non-invariance theorem of [1] guarantees that no symmetric function of the conjugates can see c_0. The present proof is consistent with this: it uses the *ordering* of the exponents A_n mod p—which residues equal 0 and which equal 1—and this datum is not symmetric under $\zeta \mapsto \zeta^t$. The weight itself is symmetric, but the assignment of weights to the two distinguished residues is not, and it is exactly this that produces the value 3 rather than an $O(\sqrt{p})$ quantity.

A correction to [1]. Lemma 2.3 shows $\prod_{j=1}^{k-1}(1+\zeta^j) = 1$ for *every* odd $k > 1$, including composite k. Consequently the "composite failure" of this product asserted in [1], and the primality-detection reading built on it, are not correct: the product cannot distinguish primes from odd composites. This does not affect any statement in the present paper, whose hypotheses are on primes $p \geq 7$; it is recorded here because the erroneous claim was flagged and corrected during formalization.

Scope. The proof uses only that $p \geq 7$ is prime (for Lemma 6.2) and odd (for the cyclotomic lemmas and the reflection $A_n + 2n + 1 \equiv A_{p-1-n}$). The primes $p = 2, 3, 5$ are genuinely excluded: at $p = 5$ the residues $a + 1$ and $p - 1$ coincide, and $p = 3$ makes 3^{-1} undefined. The constant 3 is thus a feature of primes $p \geq 7$.

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